

The University of Lahore
Department of Computer Science & IT
Final Term Examination



Spring-2024

Course Title:	Computer Organization & Assembly Language	Course Code:	CS11207	Credit Hours:	4
Course Instructors:	Mr. M. Yasin, Ms. Asmara, Mr. Faizan, Mr. Fiaz, Ms. Zarbakht	Program Name:	BSCS		
Time Allowed:	120 Minutes	Maximum Marks:	40		
Date:	20-05-24	Course Mentor:	Mr. M. Yasin Nasir		
Student's Name:		Signature:			
Reg. No:		Section:	A,B,C,D,E,F		

Important Instructions / Guidelines:

- **Remember! SOMEONE is always with you (Be Relaxed), and HE is also watching you (Be Honest)**
- Question Paper is **SELF EXPLANATORY**. Understanding the Question Paper is part of Solution Solve subjective part on the answer sheet provided separately.
- Your paper will be marked as **ZERO**, if involved in any kind of cheating.

Question # 1: (12 marks)

Given the initial values of registers and variables, find the status of the registers and variables after the execution.

a.) Initial values of registers:

AX=	4040	BX=	01234	CX=	0003	DX=	0002
BP=	0000	SI=	0000	DI=	0000	SP=	0000
ZF=	0	SF=	0	DF=	0	CF=	0

.DATA (starts at 073F:0020 h)

STR1 DB "ABCDE\$"

STR2 DB "FGHIJ\$"

. CODE

.

.

CLD

SCASB

.

.

Values after execution of given instruction:

AX=		BX=		CX=		DX=	
BP=		SI=		DI=		SP=	
ZF=		SF=		DF=		CF=	

b.) Initial values of registers:

AX=	00FB	BX=	FFFF	CX=	0003	DX=	0002
BP=	0000	SI=	0000	DI=	0000	SP=	0000
ZF=	0	SF=	0	DF=	0	CF=	0

.DATA (starts at 073F:0020 h)

. CODE

.

.

DIV BL

.

.

Values after execution of given instruction:

AX=		BX=		CX=		DX=	
BP=		SI=		DI=		SP=	
ZF=		SF=		DF=		CF=	

c.) Initial values of registers:

AX=	0ABC	BX=	ABCD	CX=	DC03	DX=	0002
BP=	0000	SI=	0000	DI=	0000	SP=	0000
ZF=	1	SF=	1	DF=	1	CF=	0

.DATA (starts at 073F:0020 h)

. CODE

.

.

SAR BX, CL

.

.

Values after execution of given instruction:

AX=		BX=		CX=		DX=	
BP=		SI=		DI=		SP=	
ZF=		SF=		DF=		CF=	

d.) Write a program to draw the box of 20x40 dimension. (Note: Your only task is to set the dimension. Forget about the other parameters.)

Question # 2:

(12 Marks)

Write an assembly language program to concatenate the two strings. You need to write two procedures, first one is STRIN and the other one is STRCONC. Main program should print the output as given in the box.

STRIN: Procedure must have initial address of string as the input. Then Ask the user to input the string of characters terminated by “Enter” key. Save the string at the specified address in memory.

STRCONC: Procedure must have initial addresses of three strings in AX, BX, and DX. Where AX and BX represent the addresses of two source strings and DX represent the address of concatenated string.

Input string1: “Final Term”

Input String2: “Exam”

Final string: “Final Term Exam”

Question # 3:**(8 marks)**

For the given code and its relevant data, find the asked values in the table.

<pre>.MODEL SMALL .STACK 100H .DATA PROMPT_1 DB 'Enter the 16-character(0/1) input : \$' PROMPT_2 DB 0DH,0AH,'The output value : \$' .CODE MAIN PROC MOV AX, @DATA MOV DS, AX LEA DX, PROMPT_1 MOV AH, 9 INT 21H XOR BX, BX MOV CX, 16 MOV AH, 1 @LOOP1: INT 21H CMP AL, 0DH JE @END AND AL, 0FH SHL BX, 1 OR BL, AL LOOP @LOOP1 @END: MOV CX, 16 XOR AX, AX @LOOP: ROL BX, 1 JNC @NEXT INC AX @NEXT: LOOP @LOOP MOV CX, AX LEA DX, PROMPT_2 MOV AH, 9 INT 21H</pre>	<pre>MOV AH, 2 CMP CL, 10 JAE @LABEL1 OR CL, 30H MOV DL, CL JMP @DISPLAY @LABEL1: MOV DL, 31H INT 21H SUB CL, 10 OR CL, 30H MOV DL, CL @DISPLAY: INT 21H MOV AH, 4CH INT 21H MAIN ENDP END MAIN</pre> Write the output here:
---	--

